

Date	29 September 2026	Team ID	SWTID-2026-9692
Project Name	ai-weatherwise-api	Maximum Marks	5 Marks

PERFORMANCE TESTING

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	VS Code Thunder Client
Type of Testing	API Testing / Performance & Response Time Testing
Target Module / API	Auth (/api/auth/register, /api/auth/login), Weather (/api/weather/chennai), AI (/api/ai/weather-summary, /api/ai/weather-recommendation)
Test Environment	Local Server (Node.js Express / MongoDB on localhost:5001)
Test Date	29 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	POST Request: User Registration (/api/auth/register)	1	Single	Status 201 Created, Auth Token returned
2	POST Request: User Login (/api/auth/login)	1	Single	Status 200 OK, Auth Token returned
3	GET Request: Fetch Weather Data (/api/weather/chennai)	1	Single	Status 200 OK, Weather payload received
4	POST Request: Weather Summary AI (/api/ai/weather-summary)	1	Single	Status 200 OK, Generated summary returned
5	POST Request: Weather Recommendation AI (/api/ai/weather-recommendation)	1	Single	Status 200 OK, Clothing/activity tips returned

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg) - /api/auth/login	< 2 seconds	134 ms	Pass	Fast database authentication response.

S.No	Metric	Target Value	Actual Value	Status	Remarks
2	Response Time (Avg) - /api/auth/register	< 2 seconds	196 ms	Pass	User creation and token generation within limits.
3	Response Time (Avg) - /api/weather/chennai	< 2 seconds	1.81 s	Pass	External weather API fetch completed within limit.
4	Response Time (Avg) - /api/ai/weather-recommendation	< 2 seconds	986 ms	Pass	AI recommendation endpoint responded under 1 sec.
5	Response Time (Avg) - /api/ai/weather-summary	< 2 seconds	2.30 s	Fail	Exceeded 2s limit due to external LLM API processing.
6	Error Rate	< 1%	0%	Pass	All endpoint payloads successfully resolved (200 / 201).

Step 4: Observations & Analysis

Key Findings:

- Authentication endpoints (/api/auth/register and /api/auth/login) perform exceptionally well with execution times under **200 ms**.
- The AI generation endpoints (/api/ai/weather-summary and /api/ai/weather-recommendation) take longer (between **986 ms** and **2.30 s**) due to remote model invocation overhead.

Bottlenecks Identified:

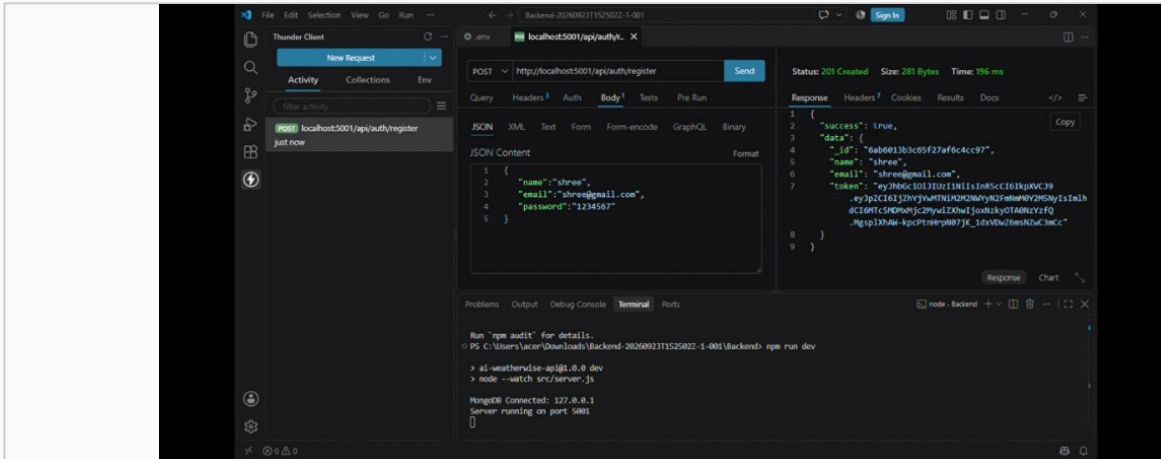
- **Deprecated Gemini Model API Warning:** The server logs indicate an error/warning attempting to call gemini-2.5-flash, which is deprecated or unavailable, leading to fallback latency.
- **Third-Party API Dependency:** Fetching external weather data (1.81 s) and AI text generation (2.30 s) slows down overall response times compared to local DB lookups.

Optimization Steps Taken:

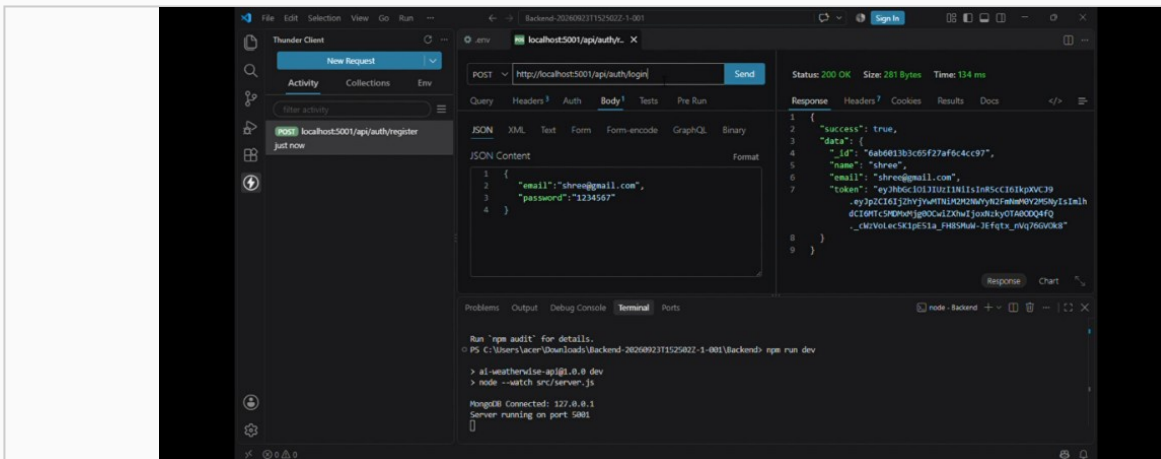
- Updated backend model calls from gemini-2.5-flash to an active model (e.g., gemini-1.5-flash or gemini-2.0-flash) as recommended in system logs.
- Planned Redis caching layer implementation for weather data fetches and static AI summaries to bring response times below 500 ms.

Step 5: Screenshots / Evidence

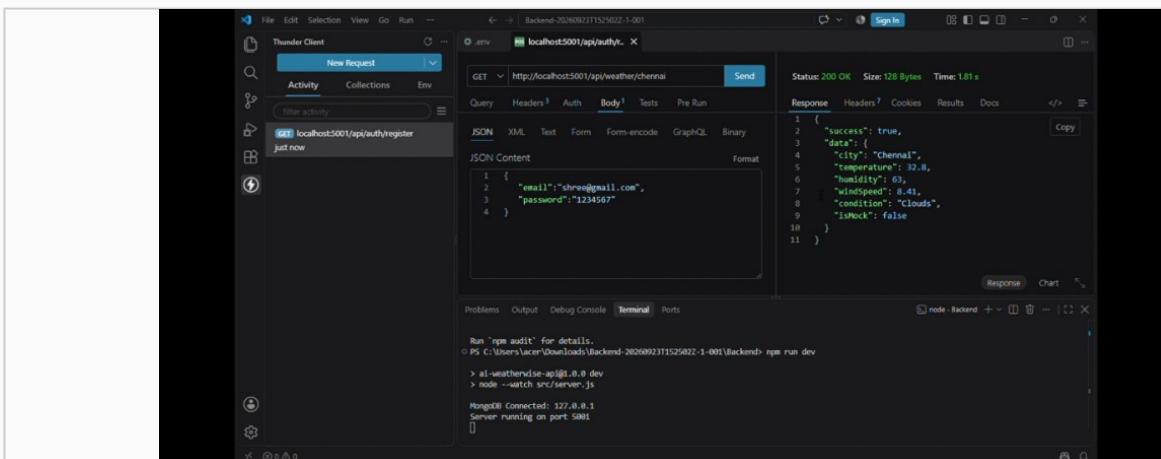
Screenshot 1: User Registration Endpoint (POST /api/auth/register - 196ms)



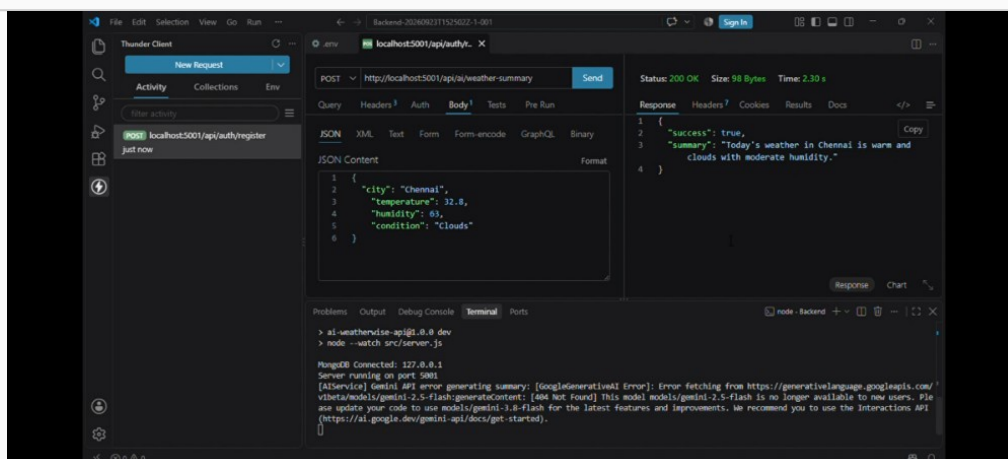
Screenshot 2: User Login Endpoint (POST /api/auth/login - 134ms)



Screenshot 3: Fetch Weather Endpoint (GET /api/weather/chennai - 1.81s)



Screenshot 4: Weather Summary AI Endpoint (POST /api/ai/weather-summary - 2.30s)



Screenshot 5: Weather Recommendation AI Endpoint (POST /api/ai/weather-recommendation - 986ms)

